

Muhammad Sarim

Software Engineer | Full-Stack Developer | AI/ML Engineer | Hybrid Graph RAG & Automation

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Education

COMSATS University Islamabad, Lahore

Bachelors of Computer Science

2022-2026

CGPA: 3.47

Technical Skills

Languages: Python, JavaScript, Dart, Java, C++

Frameworks and Libraries: React, Next.js, Tailwind CSS, Redux, FastAPI, Flutter, Django

Tools & Concepts: Git, Docker, CI/CD, Firebase, Cloudflare R2, Kubernetes and AWS (In progress)

AI/ML: Hybrid Graph RAG, RAG systems, LangChain, CNN, SVM, KNN, Transfer learning

Databases: Relational, Non-Relational, Vector databases (Qdrant), Graph databases (Neo4j)

Soft Skills

Leadership

Team Coordination

Problem Solving

Decision-making

Task prioritization

Adaptability

Attention to detail

Creative problem-solving

Initiative and self-learning

Achievements

YouTube (Non-Technical)

- Grew a YouTube Shorts channel to 630K+ subscribers with 70M+ views.
- Gained experience in marketing and creating engaging, attention-grabbing short-form content for brand and business growth by posting consistently for 300 days.

Summary

Software engineer with hands-on experience building and deploying end-to-end AI and full-stack applications. Experienced in FastAPI, React, mobile development, and hybrid multi-hop Graph RAG systems, with a strong focus on solving real-world problems through scalable and intelligent software.

Experience

Devathon

Devsinc Lahore

Worked in a small, fast-moving team to develop a responsive UI in React. Shipped a product in 6 hours.

Softec

FAST NUCES Lahore

Built a responsive UI using Next.js. Helped plan features, solve bugs, and improve user flow.

Projects

The Learning SAMPai (FYP)

- Full-stack AI tutor built on LightRAG's hybrid multi-hop Graph RAG pipeline (Neo4j knowledge graph + Qdrant vector store).
- Per-file adaptive quizzes (MCQ/TF, difficulty driven by chat history), Leitner flashcards, AI mindmaps, cross-file open-ended quizzes, real-time group chats with an @SAMPai agent, and rich announcements.
- Document ingestion pipeline (PDF, DOCX, PPTX via MinerU and Docling); scoped retrieval enforces per-classroom data isolation.
- FastAPI, React+Vite, Neo4j, Qdrant, Redis, PostgreSQL, Cloudflare R2; Docker Compose + GitHub Actions CI/CD; React Native Expo mobile app.

<https://app.learningsampai.study>

Split (Flutter Mobile app)

- A bill-splitting app to simplify real-world group expense management and save time through automated item-level cost allocation and payment tracking.
- Built with SQLite for local storage and Cloudinary for media handling, with OCR and Gemini API integration in progress to automate menu extraction.
- Future plans include user profiles, in-app payment notifications, and chat functionality for seamless event collaboration.

Pokemon Classifier (CV/ML)

- Developed image classification models for Pokémon (Pikachu and Charmander) using KNN, SVM, CNN, and transfer learning.
- Achieved progressive accuracy improvements: 57.5% (KNN), 72.5% (SVM), 90.83% (CNN), and 91.67% (transfer learning).
- Applied an iterative learning approach, advancing from basic ML to deep learning and transfer learning through experimentation and model optimization.

Research

Graph RAG vs. Classic RAG: Difficulty-Aware Quiz Generation

- Conducted a 300-question study during The Learning SAMPai project across 2 retrieval modes, 3 difficulty levels, and multi-file course content.
- Designed a two-dimensional evaluation framework measuring retrieval quality (context relevance, grounding, and difficulty alignment) and reasoning quality (Bloom level, complexity, and answerability).
- Compared Classic/Naive RAG with hybrid Graph RAG for contextual consistency and reliable difficulty-aware quiz generation; applied findings to improve the educational AI pipeline.